

Volume of prisms



1

a i Triangle ii 12 cm^2 iii 276 cm^3

b i Trapezium ii 40 cm^2 iii 240 cm^3

c i Rectangle ii 24 m^2 iii 288 m^3

d i Kite ii 16 cm^2 iii 32 cm^3

2

a 216 cm^3 b 48 cm^3 c 32 cm^3 d 217.5 cm^3

e 330 cm^3 f 84 cm^3 g 270 cm^3

3 200 cm^3

4 9600 mm^3

Volume of cylinders

5 $V = \pi r^2 h$

6

a Circle b $36\pi \text{ cm}^2$ c $360\pi \text{ cm}^3$

7

a $10\,995.6 \text{ units}^3$ b $2\,413\,754.1 \text{ units}^3$ c 785.4 cm^3 d 240.3 cm^3
 e 45.9 cm^3 f 70.7 cm^3 g 307.9 cm^3 h 1608.5 units^3
 i 883.6 cm^3 j 263.9 cm^3 k 609.5 cm^3

Volume of compound solids

8

a i A semicircle and a square ii 22.283 m^2
 iii 222.82 m^3

b

i Two rectangles

iii 68 cm^3



ii 34 cm^2

c

i Two rectangles

iii 280 cm^3

ii 40 cm^2

d

i A triangle and a rectangle

iii 412.5 m^3

ii 37.5 m^2

9

a 399 cm^3

b 1080 cm^3

c 828 cm^3

d 982.35 m^3

e 300 cm^3

f 293.52 cm^3

Unknown side given the volume

10 $a = 4 \text{ mm}$

11 $p = 6 \text{ mm}$

12 $k = 11$

13 $y = 10.1$

14 70 cm^2

15 9 cm

16 3 cm

17 9 cm

Volume of spheres

18

a 113.10 cm^3

b 4577.20 cm^3

c $113\,550.33 \text{ mm}^3$

d 33.51 cm^3



19 The student's mistake was substituting the  meter for the radius. Correct volume is 381.70 cm^3

- 20 a 817.283 mm^3 b 2572.441 cm^3 c 3.054 m^2 d 0.351 cm^2
-

Applications

21 2 m^3

- 22 a 29.69 cm^3 b 169.2 g

23 0.57 m^3

- 24 a 1335.18 cm^3 b 4005.54 cm^3 c 28 L

- 25 a 1198.5 m^3 b 299.625 m^3 c $h = 0.5875 \text{ m}$

- 26 a 8151 cm^3 b 5

- 27 a $2 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$ b 20 cm^3
c $15\,120 \text{ cm}^3$ d 756

28 $d = 29 \text{ cm}$

29 $h = 160 \text{ cm}$

30 5424.605 cm^3

31 8000 balls

32 $113\,097 \text{ cm}^3$



34 $1.646 \times 10^{11} \text{ km}^3$

35

a 32.5542 cm

b $18\,064 \text{ cm}^3$

36

a 804.2 cm^3

b 1206.4 cm^3

c 33%